



INDIAN OIL CORPORATION LIMITED
PIPELINES DIVISION HEAD OFFICE, NOIDA
STANDARD SPECIFICATION OF SCRAPER BARREL



IndianOil
(PIPELINES DIVISION)

TECHNICAL SPECIFICATIONS

FOR

SCRAPER BARREL

Standard Specification No:
IOCL-MECH-SB-081

IOCL-MECH-SB-081			Approved by	DGM (PJ-Mech)	<i>[Signature]</i> (D.N. Tripathy)
			Checked by	CPJM (Mech)	<i>[Signature]</i>
Rev.03	Date: 11.10.2022	Page 1 of 21	Prepared by	PJM (Mech)	<i>[Signature]</i> K. Anand

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PIPELINES DIVISION HEAD OFFICE, NOIDA

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Annexure-1 Inspection and Testing Plan (ITP/QAP) For Scraper Barrel.

Annexure-2 Schematic Drawing for Scraper Barrel.

Note:

1. In case of other fluid service, or design conditions, please refer section on "Additions/ Modifications/ Clarifications" detailed in Specific Requirement of Project.

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1.0.0 Abbreviation:

OWNER	INDIAN OIL CORPORATION LIMITED
SUPPLIER	BIDDER/ VENDOR/ MANUFACTURER of Equipment
TPI	Nominated THIRD PARTY INSPECTION Agency
ITP	Inspection and Test Plan
QAP	Quality Assurance Plan (ITP and QAP shall have the same meaning)

2.0.0 General:

- 2.1.0 This specification details the general requirement of Scraper Barrels suitable for launching or receiving PIGS, as well as bi-directional type scraper barrels suitable for both launching and receiving PIGS. Dimensions & Nozzle Schedules for Launching Barrels, Receiving Barrels and Bi-directional barrels are covered under this specification. This specification is to be read in conjunction with the data sheets, drawings, and specific requirement of the project.

3.0.0 Scope of Supply:

- 3.1.0 Scope of supply shall include complete mechanical design, engineering, procurement of raw materials/ bought out materials, fabrication, manufacturing, unit integration (assembly), painting, inspection, testing, documentation including certification for guarantee, preparation for shipment and shipment of all items as described in the specific requirement of the project, in accordance with the data sheet, technical specifications and all reference codes and standards of this document.
- 3.2.0 The scope shall also include providing detailed engineering of all the equipment and the components of sub-vendors. The SUPPLIER shall submit and obtain owner's approval of design documents of such equipment and components.
- 3.3.0 The scope of supply shall include supply of spares and consumables, spares for quick opening closure, etc. as indicated in the specific requirement/ datasheet.
- 3.4.0 The SUPPLIER or its sub-vendor shall perform all design calculations as indicated in this specification, reference codes and standards and the equipment datasheets. The SUPPLIER shall provide all the design inputs required for design of support/equipment foundation. The design and supply of foundation bolts for fixing of barrels are also in the scope of supply.

4.0.0 Service Conditions

- 4.1.0 The equipment including all the supplied components shall be suitably designed for service in outdoor unshaded environment with Indian climatic conditions for continuous uninterrupted service.
- 4.2.0 The design conditions and the service fluid shall be considered as per the data sheet/ specific requirement of the project.

5.0.0 Reference Regulations, Codes and Standards

- 5.1.0 The SUPPLIER shall meet or exceed the requirements of the latest edition of the following codes and standards, except as superseded herein this specification. The latest edition unless specified otherwise, of the following codes and standards shall establish the minimum standards and are a part of this specification. In case of more than one referenced standards/code, regulations and/or specifications apply to the same condition the more stringent of these shall apply:

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S. No	Title	Description
i.	ASME Section VIII Division 1	Pressure Vessel code- Rules for Construction of Pressure Vessels
ii.	ASME Section II Part D	Pressure Vessel code -Materials, Properties (Metric)
iii.	ASME Section IX	Pressure Vessel code- Welding, Brazing and Fusing Qualifications
iv.	ASME Section V	Pressure Vessel code- Non-Destructive Examination
v.	ASME B31.4	Pipeline transportation systems for liquid hydrocarbons and other liquids
vi.	ASME B31.8	Gas Transmission and Distribution Piping Systems
vii.	ASTM A 516	Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate- and Lower-Temperature Service
viii.	ASTM A 537	Standard Specification for Pressure Vessel Plates, Heat-Treated, Carbon-Manganese- Silicon Steel
ix.	API 5L	Specification for Line Pipe
x.	ASTM A106	Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
xi.	ASTM A333	Standard Specification for Seamless and Welded Steel Pipe for Low-Temperature Service and Other Applications with Required Notch Toughness
xii.	ASTM A350	Standard Specification for Carbon and Low-Alloy Steel Forgings, Requiring Notch Toughness Testing for Piping Components
xiii.	ASME B16.5	Pipe Flanges and Flanged Fittings
xiv.	ASME B16.47	Large Diameter Steel Flanges
xv.	MSS SP 44	Manufacturers Standard Society for fitting Industry - Steel Pipeline Flanges
xvi.	ASME B 16.9	Factory-Made Wrought Steel Butt Welding Fittings
xvii.	MSS SP 75	Manufacturers Standard Society for fitting Industry - Specification for high-test Wrought, Butt Welding Fittings
xviii.	MSS SP 97	Integrally Reinforced Forged Branch Outlet Fittings - Socket Welding, Threaded, and Buttwelding Ends
xix.	MSS SP 25	Standard Marking system for valves, Fittings, Flanges
xx.	ASTM A 105	Forgings, carbon steel for piping components including flanges, fittings, and valves
xxi.	ASTM A 234	Standard specification for piping fittings of wrought carbon steel and alloy steel for moderated and elevated temperatures
xxii.	ASTM A420	Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Low-Temperature Service
xxiii.	ASTM A320	Standard Specification for Alloy-Steel and Stainless-Steel Bolting for Low-Temperature Service
xxiv.	ASTM A193	Standard Specification for Alloy-Steel and Stainless-Steel Bolting for High Temperature or High-Pressure Service and Other Special Purpose Applications
xxv.	ASTM A194	Standard Specification for Carbon Steel, Alloy Steel, and Stainless-Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both

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xxvi.	ASTM A 694	Carbon and Alloy Steel Forgings for Pipe Flanges, Fittings, Valves, and Parts for High-Pressure Transmission Service
xxvii.	ASTM E 94	Standard Guide for Radiographic Examination
xxviii.	ASTM E 165	Standard Practice for Liquid Penetrant Examination for General Industry
xxix.	ASTM E 709	Standard Guide for Magnetic Particle Testing
xxx.	OISD-STD-141	Design, construction, and inspection requirements for cross country liquid hydrocarbon pipelines
xxxi.	OISD-STD-214	Cross-country LPG pipelines
xxxii.	OISD-STD-226	Natural Gas Transmission Pipelines and City Gas Distribution Networks
xxxiii.	OISD-STD-128	Inspection of Unfired Pressure Vessels
xxxiv.	IS-807	Design, Erection and Testing (Structural Portion) Of Cranes and Hoists – Code of Practice
xxxv.	IS-3177	Code of Practice for Electric Overhead Travelling Cranes and Gantry Cranes other than Steel Work Cranes
xxxvi.	IS-3832	Hand-Operated Chain Pulley Block – Specification
xxxvii.	IS-3938	Specification for Electric Wire Rope Hoists
xxxviii.	IS-3973	Code of Practice for the Selection, Installation and Maintenance of Wire Ropes
xxxix.	IS-6594	Technical Supply Conditions for Steel Wire Ropes and Strands

6.0.0 Document precedence:

6.1.0 The SUPPLIER shall notify the OWNER in case of any conflict between the specification, related data sheets and codes and standards referenced herein. The final interpretation shall rest with the Owner and confirmation in writing shall be sought by the SUPPLIER from the owner before manufacturing.

6.2.0 In case of conflict, the order of precedence shall be in the below order:

- i. Purchase order
- ii. Specific Requirement of the Project
- iii. Data Sheet
- iv. Technical Specifications
- v. Referenced Documents.
- vi. Manufacturer's standard

7.0.0 Design Requirements and Provisions

7.1.0 General:

7.1.1 The design and arrangement of the components shall be as per the GA drawing and schematic arrangement given in this specification.

7.1.2 The minimum and maximum design temperature shall be considered as specified in the data sheet. The SUPPLIER shall check and ensure the design of the barrel and its components meet the temperature conditions.

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